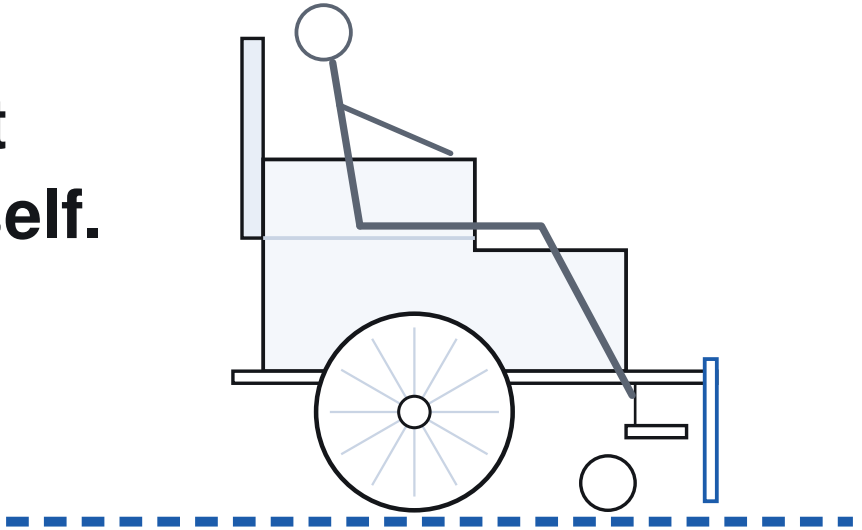
MARGADARSHAK

मार्गदर्शक — the one who shows the way

A wheelchair that takes airport passengers to their gate by itself.

It follows a line laid into the floor — so it needs no self-navigation, and costs a fraction of what that would.



the guide line it follows

Safe, inclusive and affordable mobility for women, elderly and persons with disabilities

THE PROBLEM

Assistance is promised by law and cannot be delivered by staff.

Indian airports must provide wheelchair assistance free of charge, and do not have the attendants to do it. The shortfall shows up in three specific ways.

You wait, and nobody can tell you how long

No queue you can see. No estimate. No way to know if anyone is coming.

Delivered to a gate, then forgotten

The attendant leaves. You are stranded until somebody remembers you.

The gate moves and nobody tells them

The person pushing set off committed and out of contact. Routine. Costly.

It lands hardest on elderly passengers, persons with disabilities and women travelling alone — and worse again on anyone who does not read English, because every fallback in an airport is an English sign.

HOW IT WORKS

Scan. Sit. It takes you there.

1. **The passenger scans their boarding pass.**
   Nothing typed, nothing to read, no app, no shared language needed.

2. **A laptop in the airport finds the gate.**
   The barcode names the flight. The gate is looked up live, on the premises.

3. **The chair follows the guide line.**
   No map of the world, no cloud. It obeys a speed limit and stops for obstacles.

The boarding pass does not contain the gate — and that is the argument.

It is printed at check-in; the airport assigns the gate later, and moves it. So the scan names the flight, and the gate is looked up live — a passenger can scan before a gate exists, wait in the lounge, and a chair comes back for them the moment it is announced. Nobody has to remember them.

WHAT MAKES IT DIFFERENT

Not the technology. The decisions.

Guided airport wheelchairs are deployed, not novel.
WHILL operates at around 25 airports abroad. No Indian deployment was found. Our claim is cost and context — never invention.

Cheap because of what it refuses to do

An airport is a building somebody owns and can survey. So: no self-navigation, no learned perception, no cloud. That is the whole saving.

Four languages, Konkani and Marathi first

English last — the opposite of the airport default. The build fails if a message is added in English alone. Most of why the service fails people today.

Ordered by who will miss their flight

Not who asked first. Overtaking is logged so an attendant can explain it to anyone who asks, and nobody waits more than fifteen minutes.

Nothing leaves the airport

Runs on one laptop on the premises. No cloud, no vendor account, no passenger data leaving. The booking reference is never stored.

IMPACT

What it would change, and what it costs.

~1,00,000
assistance journeys a year
across Goa's two airports (estimated)

₹4,759
the demonstrator fleet
two working chairs, built from MDF

4
languages, enforced
Konkani, Marathi, Hindi, English

0
passenger records kept
nothing leaves the building

For the passenger

A chair that comes, tells you where you are going in your own language, and does not leave you at a gate to be forgotten.

For the airport

A statutory obligation met with the staff it can actually hire, and a shift report that evidences it was met.

The same guided-path system applies unchanged to railway stations — a far larger version of the same problem.

WHERE IT STANDS

Built: the brain. Not built: the chair.

Working today

Dispatcher, routing, boarding-pass scanning, live gate changes, four languages, caravans for families, call buttons, pre-positioning, battery reserve, corridor safety, shift reporting.

271 automated tests. Runs on a laptop.

Not built

Every physical part. The chassis is designed and drawn to 4 mm MDF, the parts are costed, and nothing has been cut.

Nothing here has carried a passenger, and we make no claim that it has.

By the December exhibition

Two working demonstrator chairs on a taped guide path, running this same dispatcher: boarding pass in, gate out, the gate moved mid-journey, a call button, and a shift report at the end.

Open questions we are not pretending to have answered: regulatory classification, airport operator approval, and a native speaker's review of the four translations.